Methodology — Leakage-Free Detect → Explain → Reveal Pipeline for Text Steganography

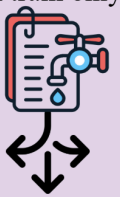
1 Cover corpora
IMDB (train / val / test)
WikiText-2 and SMS Spam
(external validation only)



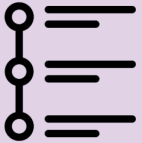
2 Stego generation
ZeroWidth, Whitespace and
Synonym channels + random
payload → 4 labelled classes



3 Leakage-free split
4 variants of a cover share
one split; TF-IDF fitted
on train only



4 Feature extraction
17 interpretable features;
TF-IDF; visible-character
coding for transformers



5 Model training
LightGBM-Lite / -TFIDF, ALBERT,
RoBERTa, Qwen-LoRA; Rule and
TextCNN baselines; multi-seed

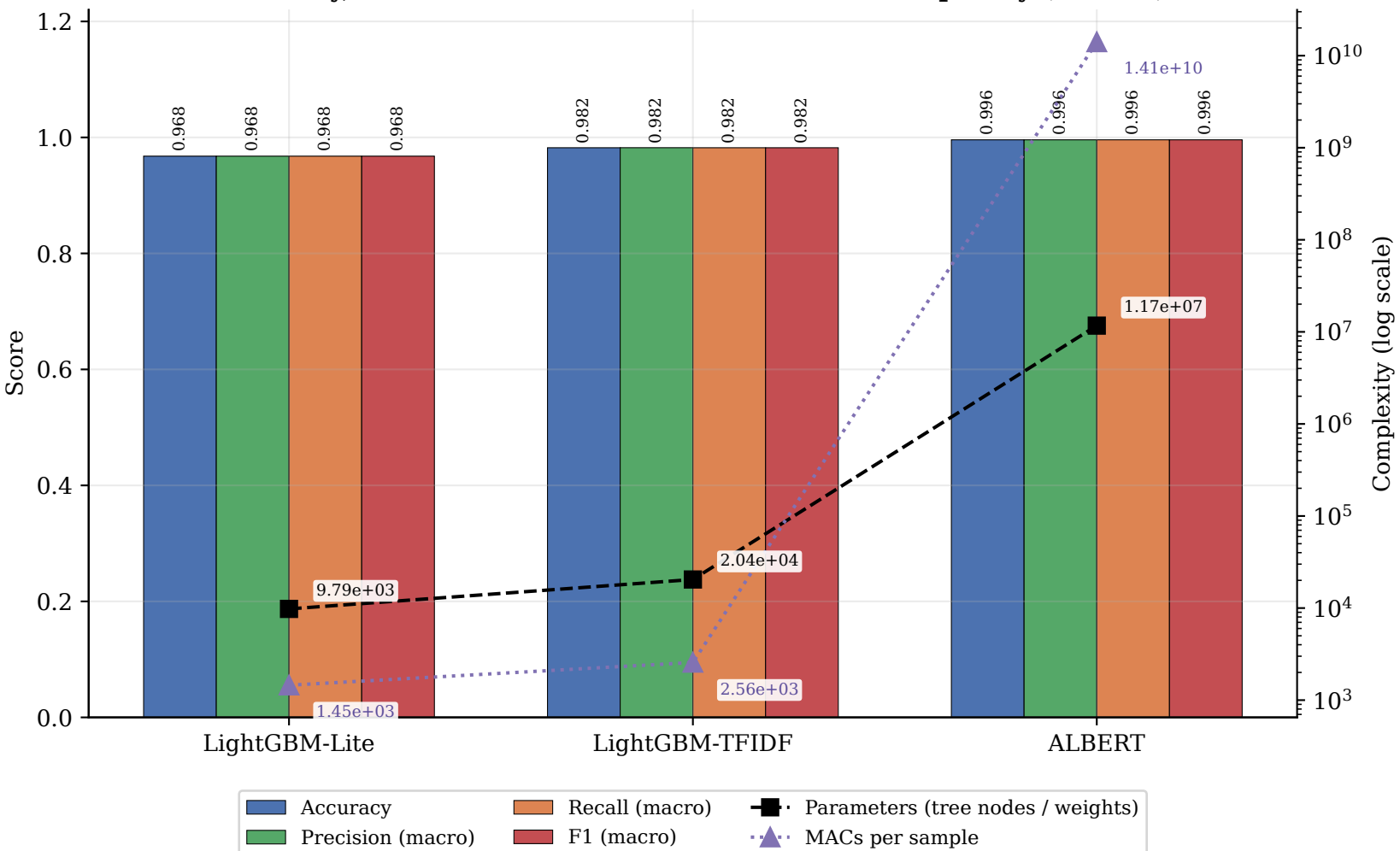


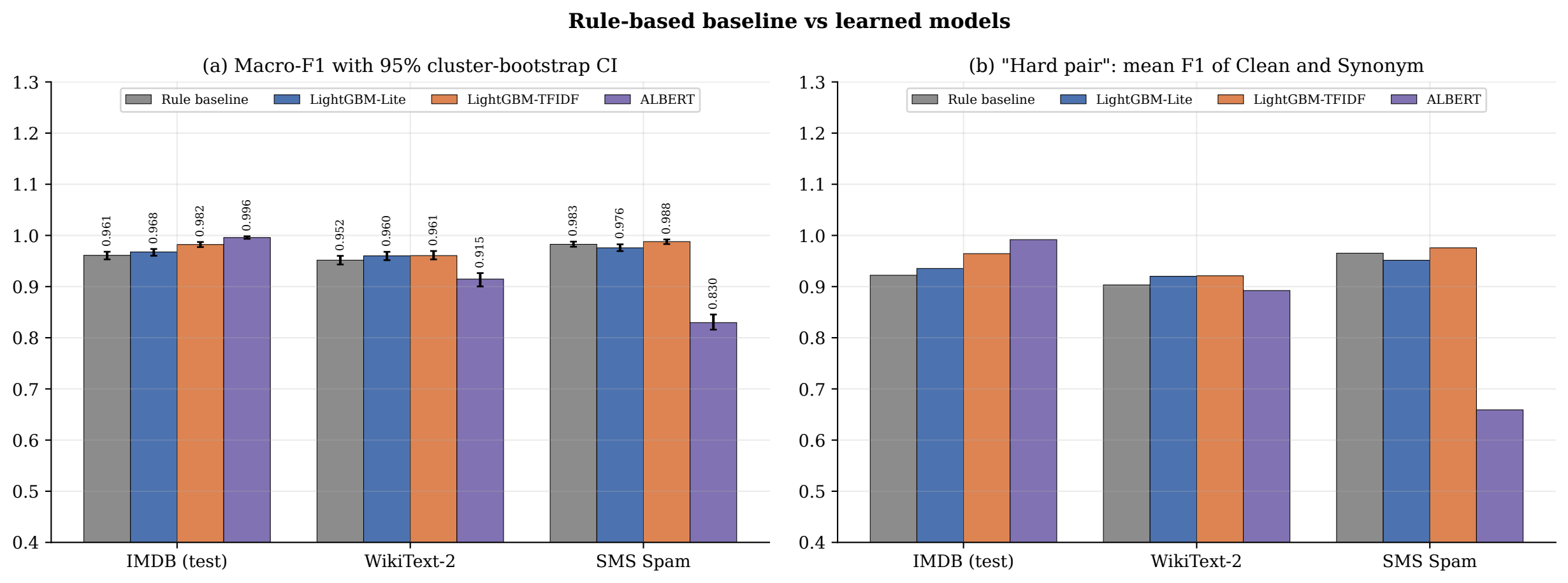
6 Evaluation
In-domain + external corpora;
1%-FPR operating points; rate,
normalisation, unseen embedders

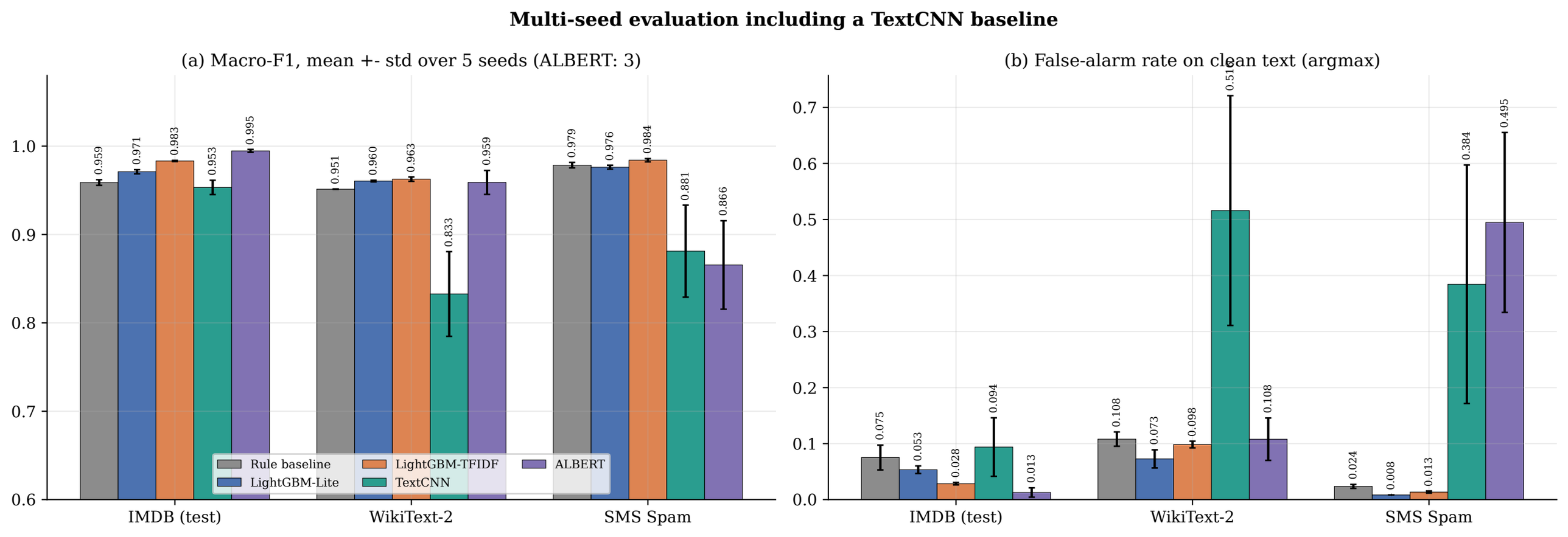


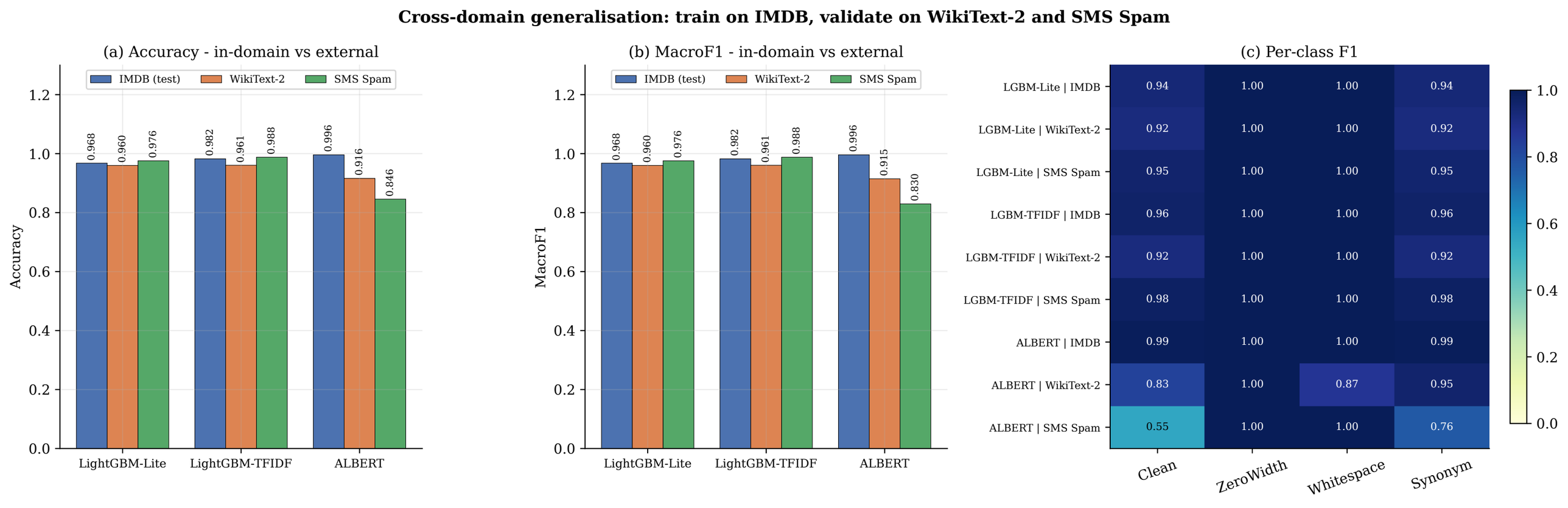
7 Explain and reveal
SHAP per alert; payload
decoding; Gradio app, models
and code released



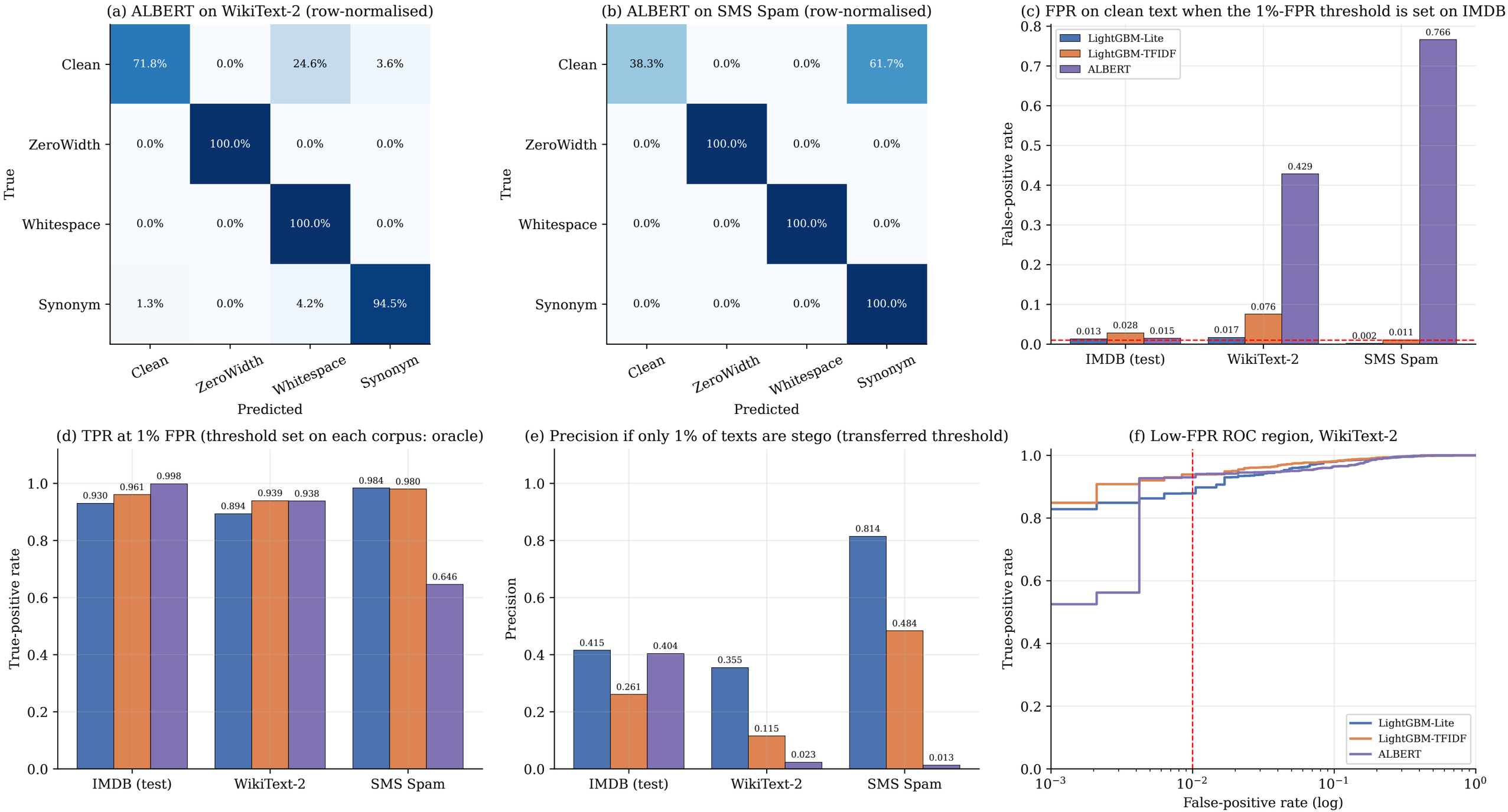

Accuracy, macro-F1/Recall/Precision vs. model complexity (test set)
Score
Complexity (log scale)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
10^{8}
10^{9}
10^{10}
0.968
0.968
0.968
0.968
0.982
0.982
0.982
0.982
0.996
0.996
0.996
0.996
9.79e+03
2.04e+04
1.17e+07
1.45e+03
2.56e+03
1.41e+10
LightGBM-Lite
LightGBM-TFIDF
ALBERT
Accuracy
Precision (macro)
Recall (macro)
F1 (macro)
Parameters (tree nodes / weights)
MACs per sample

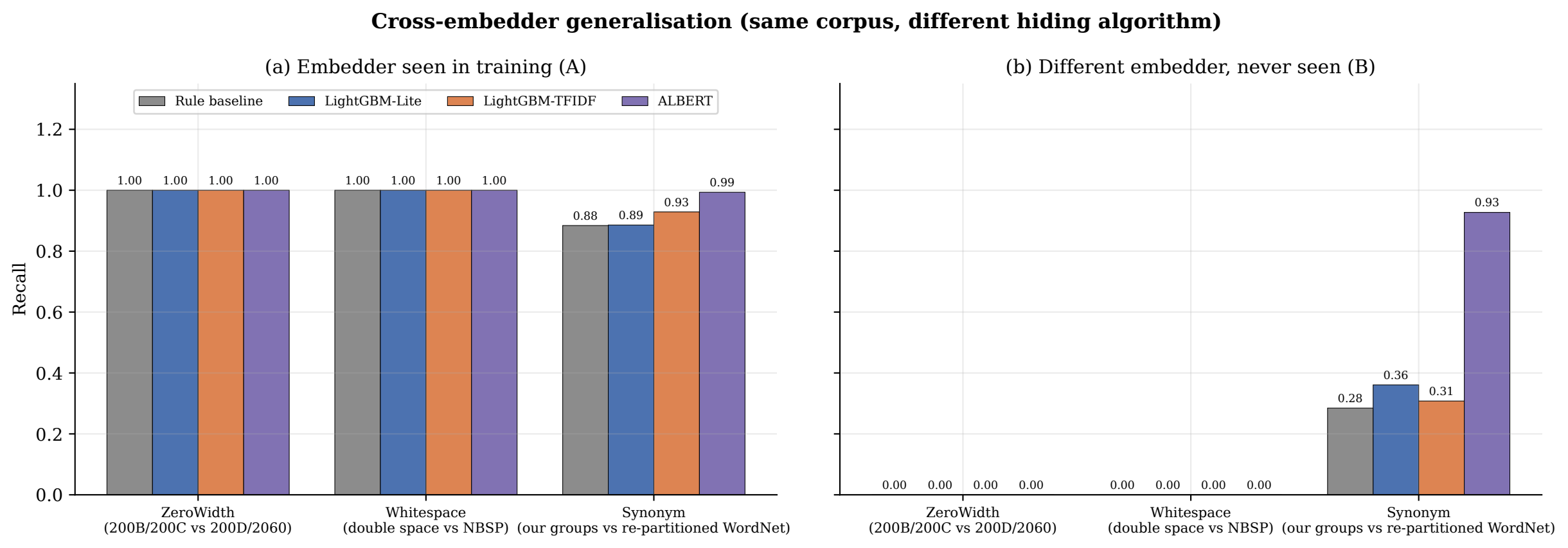

Rule-based baseline vs learned models
(a) Macro-F1 with 95% cluster-bootstrap CI
(b) "Hard pair": mean F1 of Clean and Synonym
Rule baseline
LightGBM-Lite
LightGBM-TFIDF
ALBERT
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.961
0.968
0.982
0.996
0.952
0.960
0.961
0.915
0.983
0.976
0.988
0.830
IMDB (test)
WikiText-2
SMS Spam


Multi-seed evaluation including a TextCNN baseline
(a) Macro-F1, mean +- std over 5 seeds (ALBERT: 3)
(b) False-alarm rate on clean text (argmax)
Rule baseline
LightGBM-Lite
LightGBM-TFIDF
TextCNN
ALBERT
IMDB (test)
WikiText-2
SMS Spam
0.959
0.971
0.983
0.953
0.995
0.951
0.960
0.963
0.833
0.959
0.979
0.976
0.984
0.881
0.866
0.075
0.053
0.028
0.094
0.013
0.108
0.073
0.098
0.516
0.108
0.024
0.008
0.013
0.384
0.495

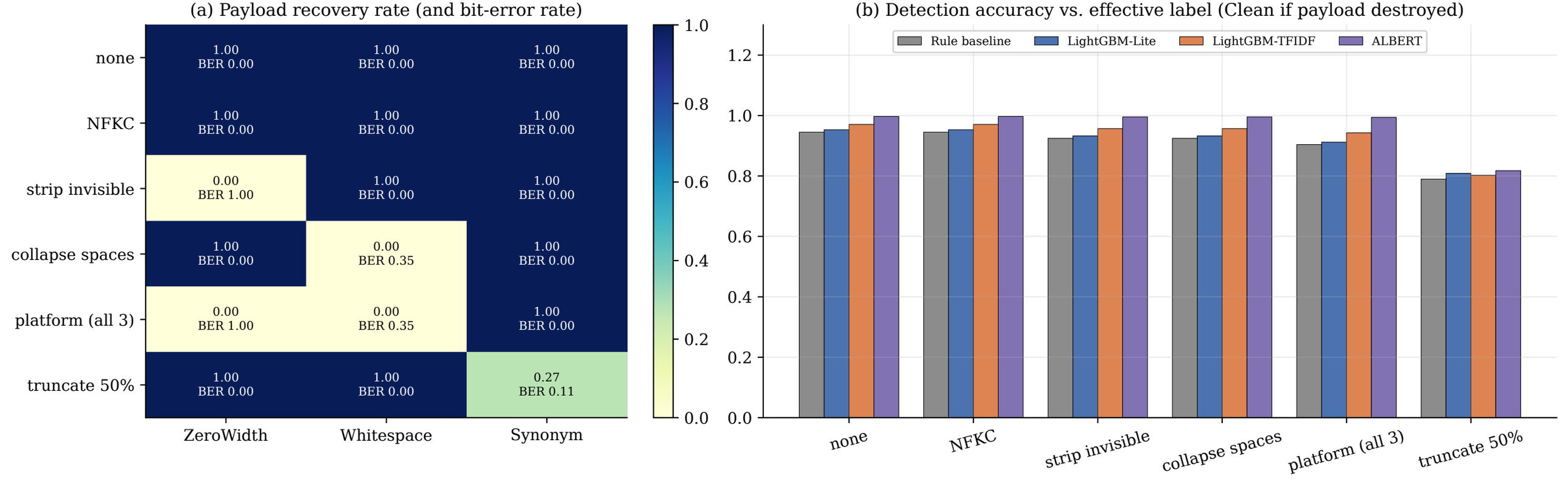

Cross-domain generalisation: train on IMDB, validate on WikiText-2 and SMS Spam
(a) Accuracy - in-domain vs external
IMDB (test)
WikiText-2
SMS Spam
Accuracy
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.968
0.960
0.976
0.982
0.961
0.988
0.996
0.916
0.846
LightGBM-Lite
LightGBM-TFIDF
ALBERT
(b) MacroF1 - in-domain vs external
IMDB (test)
WikiText-2
SMS Spam
MacroF1
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.968
0.960
0.976
0.982
0.961
0.988
0.996
0.915
0.830
LightGBM-Lite
LightGBM-TFIDF
ALBERT
(c) Per-class F1
LGBM-Lite | IMDB
LGBM-Lite | WikiText-2
LGBM-Lite | SMS Spam
LGBM-TFIDF | IMDB
LGBM-TFIDF | WikiText-2
LGBM-TFIDF | SMS Spam
ALBERT | IMDB
ALBERT | WikiText-2
ALBERT | SMS Spam
0.94 1.00 1.00 0.94
0.92 1.00 1.00 0.92
0.95 1.00 1.00 0.95
0.96 1.00 1.00 0.96
0.92 1.00 1.00 0.92
0.98 1.00 1.00 0.98
0.99 1.00 1.00 0.99
0.83 1.00 0.87 0.95
0.55 1.00 1.00 0.76
Clean
ZeroWidth
Whitespace
Synonym
1.0
0.8
0.6
0.4
0.2
0.0

False-alarm analysis: stego-vs-clean detection at operating points that matter



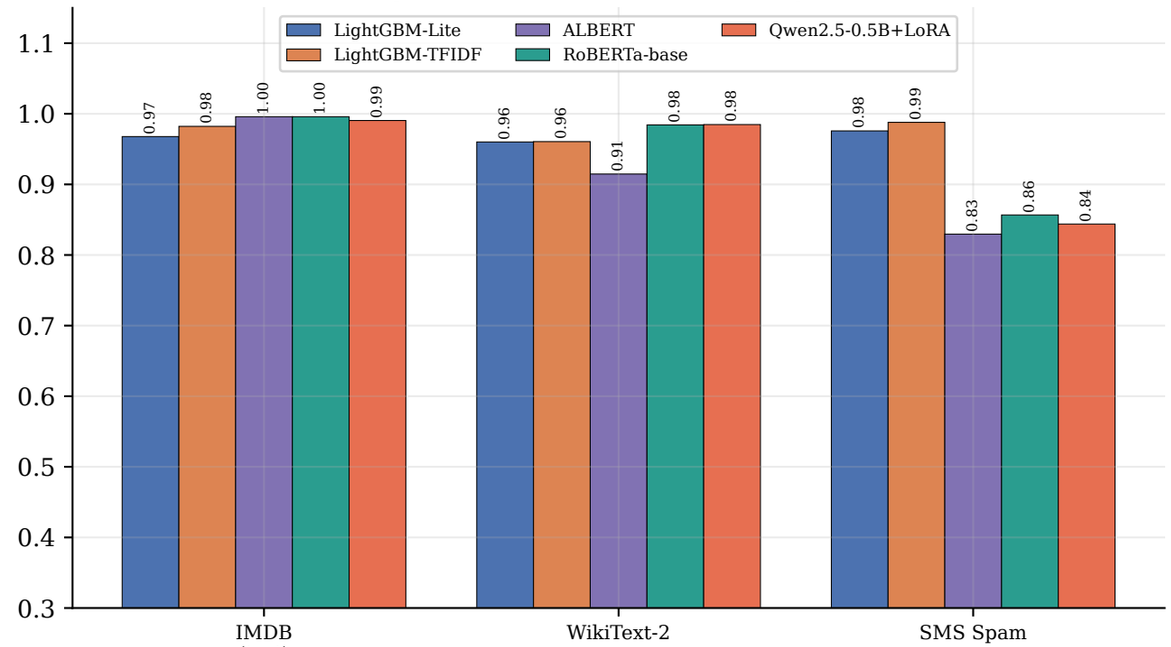

Cross-embedder generalisation (same corpus, different hiding algorithm)
(a) Embedder seen in training (A)
(b) Different embedder, never seen (B)
Rule baseline
LightGBM-Lite
LightGBM-TFIDF
ALBERT
Recall
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.00 1.00 1.00 1.00
1.00 1.00 1.00 1.00
0.88 0.89 0.93 0.99
0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00
0.28 0.36 0.31 0.93
ZeroWidth
(200B/200C vs 200D/2060)
Whitespace
(double space vs NBSP)
Synonym
(our groups vs re-partitioned WordNet)

Robustness to Unicode normalisation, whitespace clean-up, invisible-char stripping and truncation

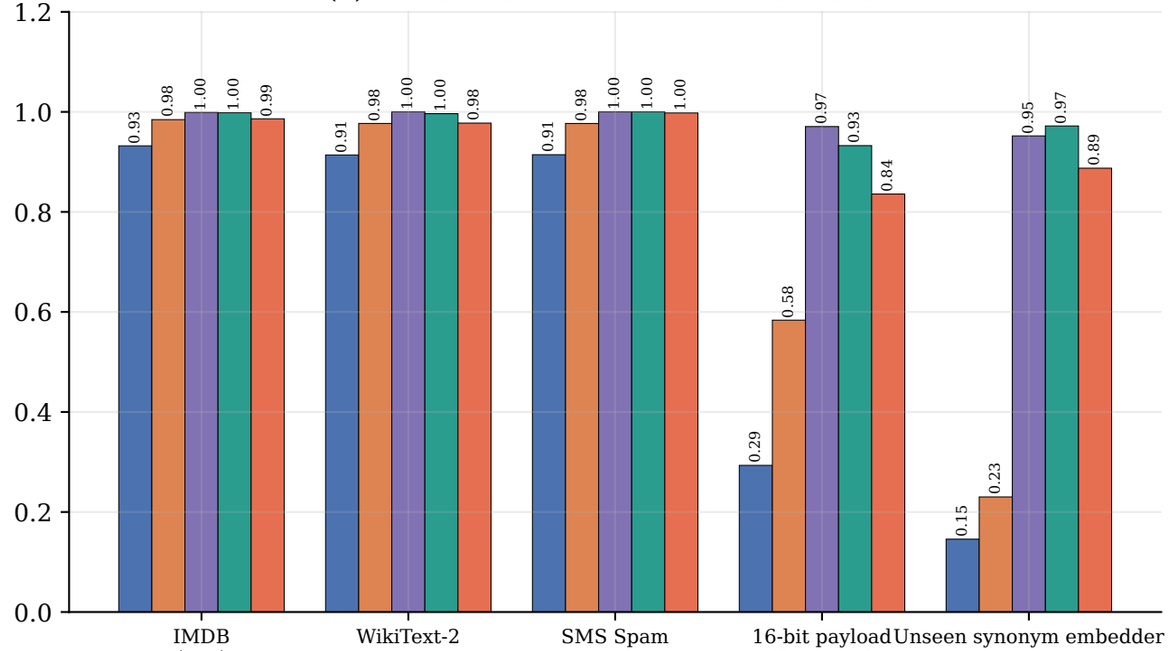


Larger-encoder baseline (RoBERTa-base) under the same protocol

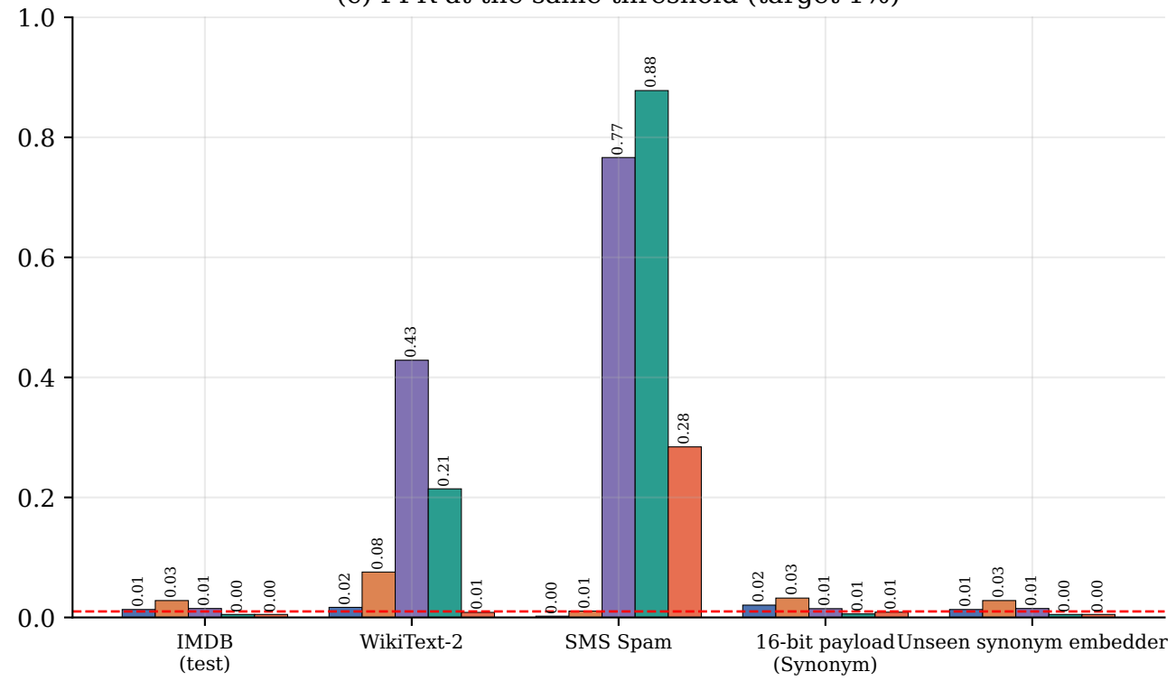
(a) Macro-F1 (single seed; see the multi-seed figure for seed spread)



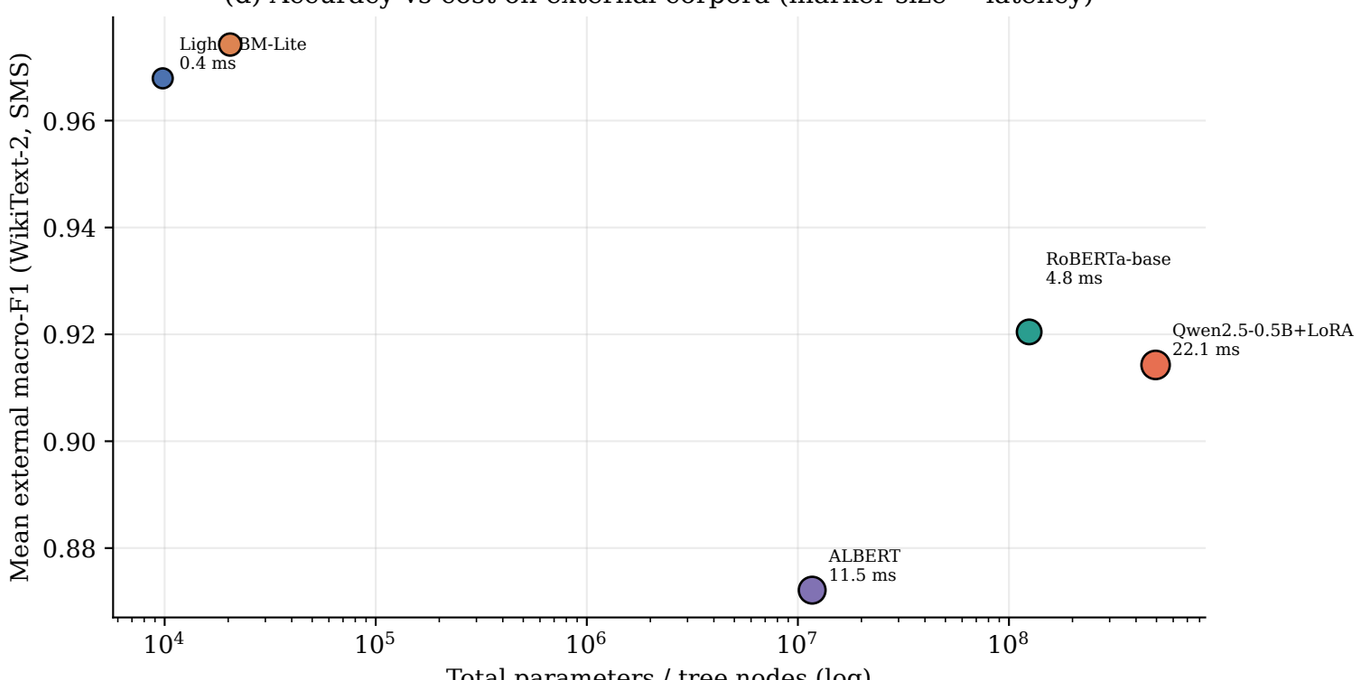
(b) TPR at the IMDB-val 1%-FPR threshold

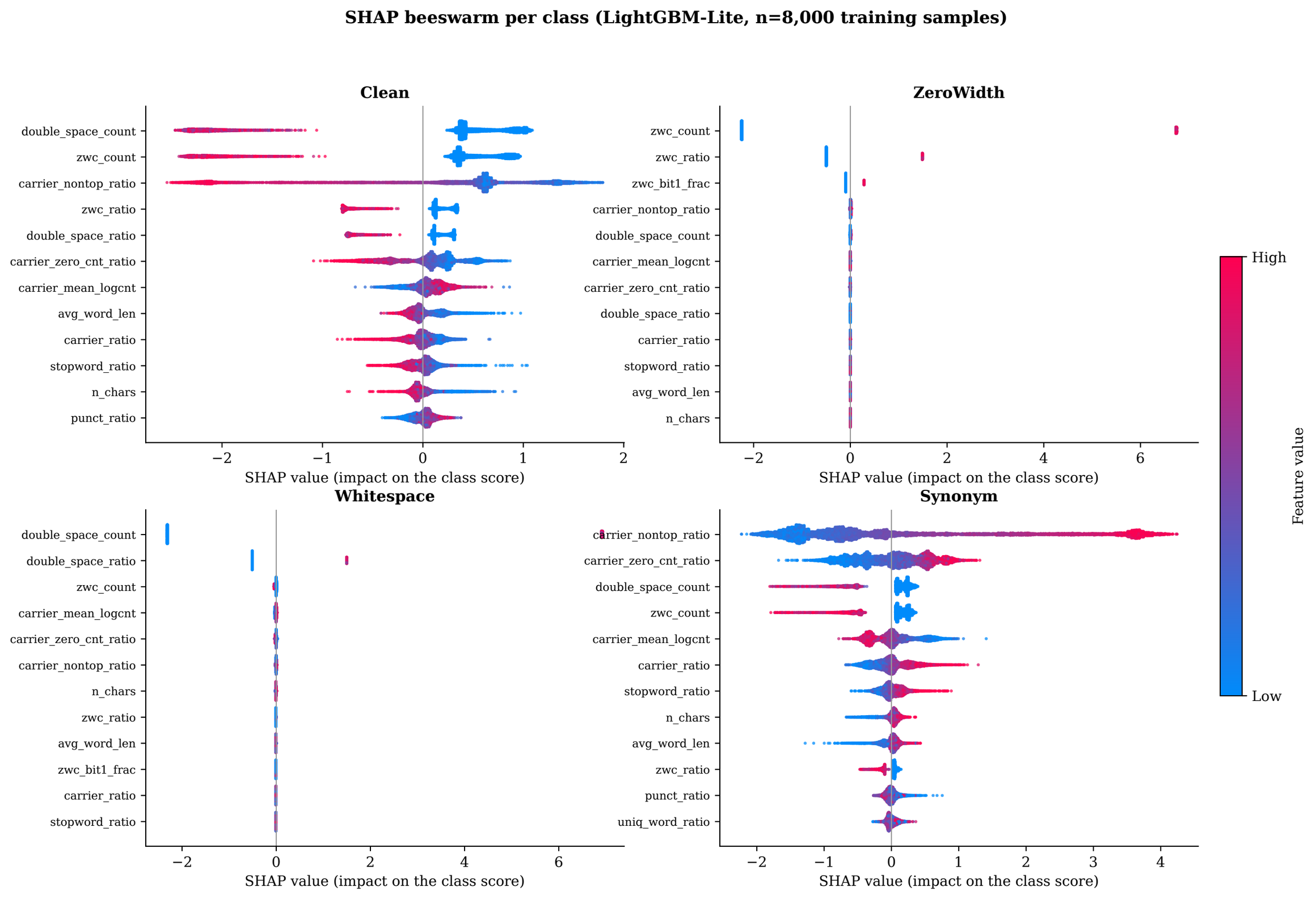


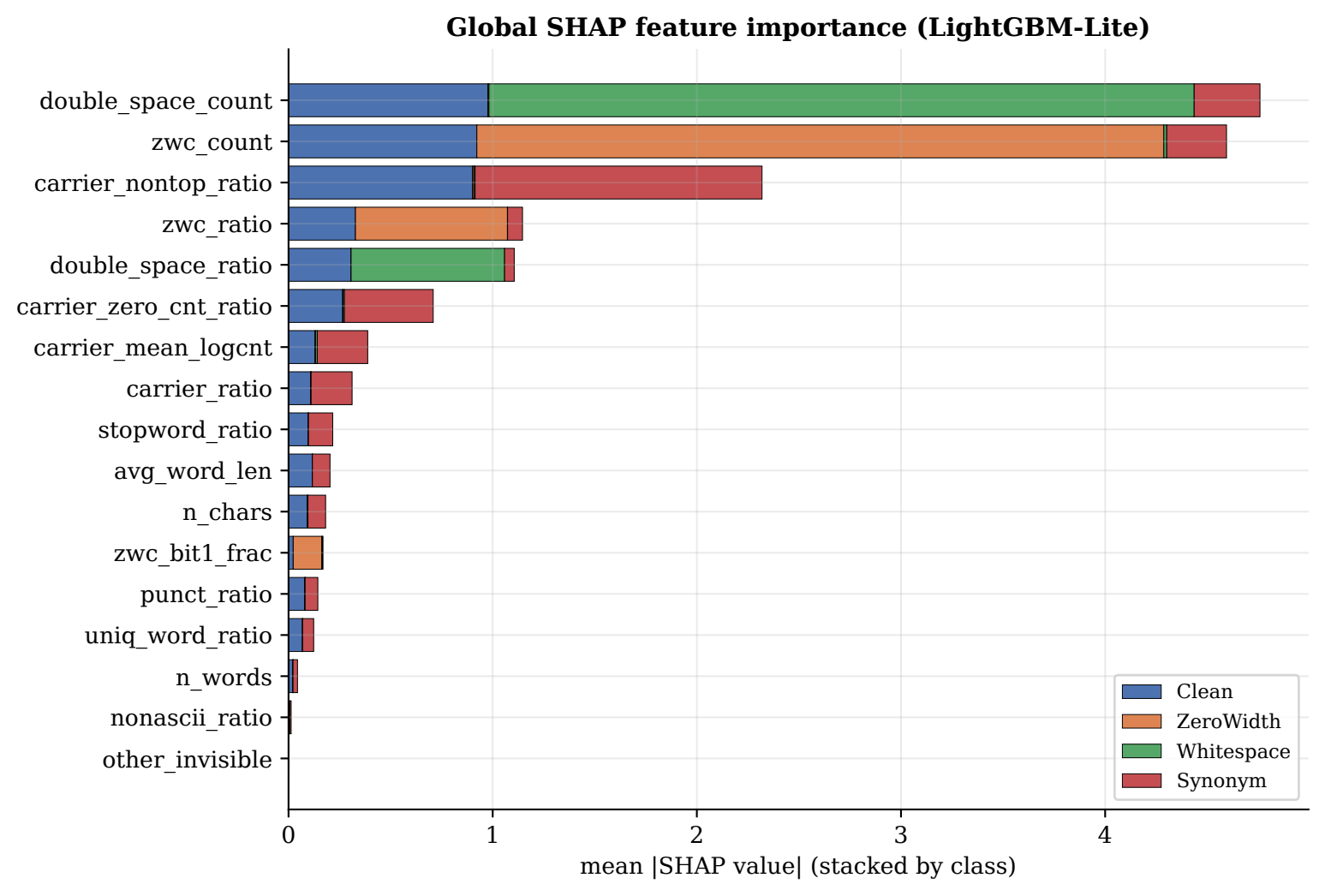
(c) FPR at the same threshold (target 1%)



(d) Accuracy-vs-cost on external corpora (marker size ~ latency)




SHAP beeswarm per class (LightGBM-Lite, n=8,000 training samples)
Clean
double_space_count
zwc_count
carrier_nontop_ratio
zwc_ratio
double_space_ratio
carrier_zero_cnt_ratio
carrier_mean_logcnt
avg_word_len
carrier_ratio
stopword_ratio
n_chars
punct_ratio
−2
−1
0
1
2
SHAP value (impact on the class score)
ZeroWidth
zwc_count
zwc_ratio
zwc_bit1_frac
carrier_nontop_ratio
double_space_count
carrier_mean_logcnt
carrier_zero_cnt_ratio
double_space_ratio
carrier_ratio
stopword_ratio
avg_word_len
n_chars
−2
0
2
4
6
SHAP value (impact on the class score)
Whitespace
double_space_count
double_space_ratio
zwc_count
carrier_mean_logcnt
carrier_zero_cnt_ratio
carrier_nontop_ratio
n_chars
zwc_ratio
avg_word_len
zwc_bit1_frac
carrier_ratio
stopword_ratio
−2
0
2
4
6
SHAP value (impact on the class score)
Synonym
carrier_nontop_ratio
carrier_zero_cnt_ratio
double_space_count
zwc_count
carrier_mean_logcnt
carrier_ratio
stopword_ratio
n_chars
avg_word_len
zwc_ratio
punct_ratio
uniq_word_ratio
−2
−1
0
1
2
3
4
SHAP value (impact on the class score)
High
Low
Feature value


Global SHAP feature importance (LightGBM-Lite)
double_space_count
zwc_count
carrier_nontop_ratio
zwc_ratio
double_space_ratio
carrier_zero_cnt_ratio
carrier_mean_logcnt
carrier_ratio
stopword_ratio
avg_word_len
n_chars
zwc_bit1_frac
punct_ratio
uniq_word_ratio
n_words
nonascii_ratio
other_invisible
Clean
ZeroWidth
Whitespace
Synonym
0
1
2
3
4
mean |SHAP value| (stacked by class)

Reveal evaluation: payload recovery and false alarms

